Creativity from Surprise: Bridging the Gap Between Fashion Designers' Inspiration Work and AI Creative Support Tools

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Abstract

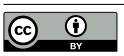
Advances in Generative AI (GenAI) enable unexpected creation in visual images. In fashion design, this capability has intensified demand for creativity support tools where fast-paced trends challenge fixation and drive exploration of novel creative directions. While prior work has explored interfaces that align designer intent with GenAI outputs, we still lack an empirical understanding of how fashion designers define, seek, and utilize AI-generated surprise as a valuable resource and actionable design direction rather than random noise. We address this gap through a qualitative study combining semi-structured interviews with 20 fashion professionals and a design workshop with 12 graduate students. We conceptualized surprise as a strategy that can be designed into GenAI-powered visualization tools to support traceable exploration, contextual grounding, and controllable variation across ideation stages. This work (1) reframes surprise as a designable mechanism or resource for co-creative interaction, (2) provides empirical insights into how fashion designers can utilize AI-generated surprise in the early stage of design, and (3) translates these insights into actionable guidance for building GenAI-driven visualization tools for fashion and related creative domains from a human-centered AI perspective.

CCS Concepts

• **Human-centered computing** → *Interaction design theory, concepts and paradigms*; Web-based interaction; **User interface design**.

Keywords

Surprise, Inspiration, Fashion design, Generative AI, Human-AI Co-creation, Creativity Support Tools



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1 Introduction

Rei Kawakubo, founder of Comme des Garçons, one of the well-known avant-garde fashion brands, notes: *"If you had the definition of what is beautiful, I would never be able to find anything. Because often the beauty comes from the unexpected."* [137] As she suggested, in fashion design, the unexpectedness or surprise is not an anomaly but a driver of creative exploration. Although surprise is particularly consequential in early-stage ideation, yet difficult to operationalize in practice. The process involves divergent thinking, visual identity formation, and narrative building [54]. This makes the ability to capture and interpret unexpectedness, or surprise, especially important for designers, who often struggle with fixation, stylistic repetition, and reliance on familiar visual patterns. Designers therefore seek a way to harness surprise: generative outcomes that meaningfully stretch a concept while remaining anchored to brand identity, market context, and material constraints. Prior work suggests that designers can break out of repetition, explore new directions, and evoke emotional resonance by seeking surprising stimuli such as visual, material, or conceptual [2, 18, 58, 114, 125, 146, 147]. Such moments of surprise can become turning points that shape new design identities and visual narratives [68, 72]. We draw on cognitive models to define surprise as a discrepancy between expectations and observed reality [3, 25, 110]. We focus on the conditions that give rise to surprise, which helps clarify what triggers it in the early-stage fashion ideation.

With the rise of Generative AI (GenAI) tools, they can occasionally produce unconventional or surprising combinations. However, their outputs also reflect the social, aesthetic, and geographic biases embedded in their training data, which can reinforce dominant styles rather than diversify them [8, 109, 115]. Even though fashion

designers report that GenAI can accelerate ideation, reveal overlooked references, and produce unexpected juxtapositions, they also note that surprise is inconsistent, incoherent, or misaligned with their creative intentions[78, 101, 111, 158]. Therefore, current GenAI systems frequently struggle to maintain brand consistency, interpret professional design language, or support long-term iterative workflows [60]. As a result, surprising outputs may fail to translate into usable inspiration.

In this paper, we focus on visual-conceptual surprise in image-based GenAI outputs during early-stage fashion ideation. We aim to explore fashion designers' interpretations and needs to guide these tools, enabling them to better support strategic, controllable, and context-aware co-creation.

Specifically, we address the research question "How should strategic surprise be conceptualized and supported in GenAI-powered visualization tools for early-stage fashion ideation?" We break it into three sub-questions as follows:

- **RQ1:** How do professional fashion designers conceptualize and strategically utilize surprise in early-stage ideation?
- **RQ2:** What are the capabilities and limitations of current GenAI-powered visualization tools in supporting surprise-driven creativity in fashion design?
- **RQ3:** What system features are needed to enable user-aligned, controllable, and context-aware surprise in GenAI-powered visualization tools?

To answer our questions, we ground our argument in a two-part qualitative study:

(1) Semi-structured interviews with 20 professional fashion designers revealed how surprise influences creative judgment and how designers negotiate initiative when working with GenAI. We frame surprise as an adjustable design parameter that can be tuned, interpreted, and strategically deployed in early fashion ideation.
(2) A design workshop with 12 graduate design students explored system features that could better support user-aligned, controllable, and context-aware surprise. These findings informed a four-stage framework: setting expectations, supporting interaction, recovering from errors, and improving with use that supports long-term co-creative engagement.

Based on our findings from the interviews and workshop, we offer two primary contributions:

(1) We identify how designers use surprise both as an entry point and a strategy, and document the current limitations of GenAI tools, including a lack of interpretability, control, and contextual sensitivity. We also surface how designers reclaim initiative from AI and distinguish between Internal Surprise (cognitive disruption) and External Surprise (market or cultural insight).
(2) We propose a methodological blueprint outlining how GenAI-powered visualization tools could support strategic surprise that supports dialogic co-creation between designers and GenAI tools. This includes actionable system features, such as prompt hierarchies, real-time feedback, zoom-out views, and long-term memory, that enable designers to co-direct and refine surprising outputs across iterative stages of design.

By reframing GenAI-supported surprise as a GenAI-powered visualization tool capable of producing meaningful surprises, we expand the design space of co-creative AI. Rather than treating surprise as incidental novelty, we argue it should be interpretable, aligned with user goals, and embedded in a human-centered workflow to genuinely support professional creativity. Although our empirical case is situated in fashion design, the study speaks to broader HCI questions about how practitioners in visual domains engage AI tools during creative work to seek unexpected inspiration. Fashion provides a concrete and relatable context, but the methodological blueprint and interaction patterns we identify are transferable across domains, making the contribution relevant to the wider HCI community.

2 Related Work

2.1 The Foundation of Surprise in Fashion Creativity

2.1.1 Early-stage fashion design process. The fashion design process follows a structured workflow including research, development, prototyping, approval, and production [27, 32, 53, 70, 96]. Early-stage fashion design, often referred to as the ideation or concept development phase, focuses on exploring divergent creative directions before decisions about construction, materials, and production are made [24, 92]. This stage typically precedes sketching, prototyping, and technical development, and is characterized by broad visual research, theme building, and where substantial visual experiments with stimulating references are needed [18, 37]. Designers may either gather broad references to spark inspiration or investigate a specific direction in depth [91]. Fashion designers transform, combine, and adapt elements of existing designs to formulate novel ideas [143]. Any perceptual experience or memory can give a designer an idea [24], including runway archives, street photography, cultural artifacts, and personal experiences.

2.1.2 Conceptualization of surprise in fashion design. Surprise is considered one of the essential and distinct characteristics of creative design [86]. Because surprise has been shown to disrupt fixation, reorient perceptual expectations, and activate emotional engagement, key factors in early-stage ideation [36, 68, 107]. Fashion design is uniquely characterized by applying visual surprise, including unexpected visual stimuli, combinations, or representations, to offer unexpected insights and creative discoveries in the ideation stage [130].

In cognitive psychology, surprise is viewed as a discrete neutral emotional response to violated expectations, a discrepancy between active schemas and incoming information [93, 98, 99, 112]. In design studies, surprise is often linked to novelty or is defined in a way that incorporates unexpectedness [6, 34, 87, 120, 139]. We define surprise as the result of encountering something suddenly or unexpectedly, a discrepancy between an expectation and an observed reality [3]. We have compiled definitions for each term and how they are usually manifested in AI-generated images to Table 1 differentiate our focus, *Surprise*, from other interpretations.

We distinguish three roles of surprise as an entry point, a strategy, and a practical approach in professional fashion practice. First,

Concept	Working Definition	Manifestation in AI-Generated Images
Random	Pure noise without pattern or semantic meaning; outputs that lack any coherent relationship to inputs or domain knowledge, resulting from training failures, hallucinations, or complete model incomprehension [17, 55, 89, 149].	• Semantic nonsense: Text labels in unknown languages, garbled typography • Anatomical impossibilities: Six-fingered hands, faces with distorted features • Physics violations: Fabrics defying gravity without stylistic intent • Contextual incoherence: Summer coats on beach scenes, evening gowns in industrial settings • Training artifacts: Watermarks, stock photo remnants, dataset biases
Unexpected	An outcome that deviates from what was anticipated based on prior experience or contextual norms, but not necessarily violating deep-seated schemas [40, 48, 52, 86, 100, 106].	• Color combinations outside conventional palettes • Fabric textures that differ from prompt specifications • Style blending (e.g., Victorian + streetwear) • Proportions or silhouettes slightly modified from input • Contextual elements (background, setting) not specified
Unpredictable	Characterized by high variability and low controllability in outcomes, where the same inputs yield inconsistent results due to stochastic processes or insufficient parameter specification [149, 152].	• Inconsistent outputs from identical prompts across generations • Random artifacts (distorted faces, malformed hands, impossible geometries) • Semantic drift between prompt and output • Style instability across iterative refinement • Cultural misinterpretation: Banana as metaphor for anxiety lost in translation
Surprise	A discrete emotional response to a discrepancy between active schemas and incoming information, involving expectation violation that triggers curiosity and reorientation [93, 98, 99, 112].	• Garment construction logic beyond designer's expertise (e.g., impossible sleeve joins, novel draping techniques), *referred as Internal Surprise in this study* • Market-relevant insights (e.g., emerging demographics, underserved niches), *referred as External Surprise in this study* • Visual metaphors reinterpreted across cultural contexts • Material combinations violating physical properties

Table 1: Working definitions of uncertainty-related concepts and how they may manifest in AI-generated images.

surprise serves as an inspirational stimulus that fashion designers routinely draw from a broad spectrum of sources, including couture archives, streetwear, historical garments, and digital trend forecasting platforms [127]. Fashion designers synthesize a multitude of stimuli into coherent narratives, often guided by tacit cultural fluency and brand-specific style grammars [92]. Surprise is also a "strategy," used not only to produce distinct visual language but also to create iconic aesthetic moments that linger in memory and define a designer's voice [130, 131]. During the idea stage, designers use their background experiences and skills, as well as stimuli they might have access to [37], including various strategies, such as visual-tactile incongruity and unexpected material combinations, to elicit surprise and yield unforeseen but impactful creative directions [38, 107]. Additionally, as an operational practice, surprise represents a dual imperative of artistic autonomy and commercial viability. Designers are expected to generate novelty without alienating consumers or violating brand identity [14], as well as considering utility, fit, value, and ease of care [91]. This process not only seeks innovation but does so within tight commercial constraints, aesthetic novelty must align with recognizability, wearability, and market receptivity. This tension also shapes decisions such as producing limited ranges to maintain exclusivity [91].

Taken together, fashion design is a complex process [91] that involves distinct personal design vocabularies, developed through exposure to inspiring stimuli [41, 72]. Current studies that include surprise as either a prerequisite for producing creative products or as a separate aspect of creativity are still lacking in the design literature [5]. This foregrounds our first research question RQ1: *"How do professional fashion designers conceptualize and strategically utilize surprise in early-stage ideation?"* in our research.

2.2 The Role of GenAI in Facilitating Creative Ideation

2.2.1 Opportunities and limitations of GenAI in early-stage fashion ideation. GenAI can be a powerful tool for developing fashion design ideas, as one of its function, image generation, perfectly matches the needs of fashion inspiration. This is reflected in designers often drawing on garment photographs during the ideation stage [127]. GenAI technologies are introducing computational mechanisms for visual ideation, image synthesis, and prompt-driven content generation. These technologies are especially relevant to early-stage workflows, where designers gather broad reference material to focus ideation [91]. Tools like DALL·E 2, Midjourney, and Stable Diffusion enable designers to generate novel visual content from text inputs, often yielding results that provoke user surprise

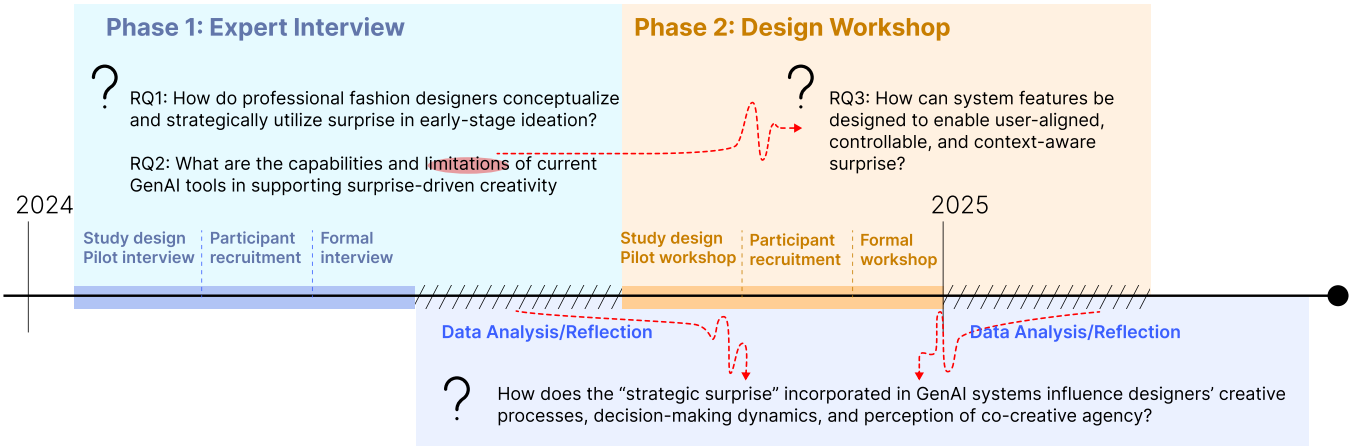


Figure 1: Timeline of two-phase research process. Phase 1: Expert Interview and Phase 2: Design Workshop.

by exceeding aesthetic or semantic expectations [64, 79, 108]. In fashion venues, tools like StyleMe [145], and Fashion-Sketcher [30] assist designers in creating and refining sketches from input images by utilizing the style transfer capabilities. Generative.fashion allows users to create designs and explore variations via graphical interfaces [19, 20]. These tools function partly by sampling from high-dimensional latent spaces, allowing for outputs that defy common compositional norms, as demonstrated by Lin et al.'s robot Cobbie that elicited inspiration through unplanned sketching trajectories [79].

In early-stage fashion ideation, user studies indicate that the efficacy of these systems is double-edged. While users often appreciate serendipitous outputs, they also express concern over the lack of control and interpretability of such surprises, particularly in high-stakes or commercially sensitive design tasks [12, 28, 95]. GenAI-driven surprise can introduce uncertainties around data provenance, copyright boundaries, and culturally biased representations, which in turn shape how designers interpret and trust AI-generated outputs [8, 109, 115]. While AI automates certain dimensions of creativity, it also risks flattening it by replacing the reflective processes that underpin meaning-making in design [50]. Furthermore, GenAI enhances individual creativity, but it may reduce the overall diversity of creative ecosystems [23]. Finally, most GenAI systems lack feedback architectures that would allow users to iteratively direct or refine the kind of surprise introduced, resulting in frustration or design fixation [56, 58, 59]. Tools like Sentient Sketchbook represent early attempts to embed co-creative dynamics, but lack the fine-grained user control necessary for domain-specific design challenges such as fashion [77]. Overall, the absence of user-controlled parameters, limited cultural contextualization, and failure to support collection-level coherence are recurring critiques. This foregrounds our second research question RQ2: *"What are the limitations of current GenAI-powered visualization tools in supporting surprise-driven creativity?"* in our research.

2.2.2 Toward strategic and controllable surprise in GenAI tools. Current research that attempts to use GenAI to enhance design creativity, overlooks a core value of GenAI: the unexpected surprises that naturally arise during image generation, which are crucial for generating creativity [41, 72]. As mentioned in the Section 2.1.2, the surprise we discuss involves the various conditions that evoke it, revealing the deviation between the outcome and the expectation, the manner in which it occurred, and the value it generates [3]. This implies that surprise can be treated as a parameter that designers may want to increase, constrain, or redirect through interface design, depending on context. By advocating for GenAI-powered visualization tools that deliver *strategic surprise* [130, 131], outputs that are both novel and semantically grounded in user intent [44]. This foregrounds our last research question RQ3: *"What system features are needed to enable user-aligned, controllable, and context-aware surprise in GenAI-powered visualization tools?"* in our research.

3 Study Design

We adopted a two-phase qualitative research design to investigate how fashion designers conceptualize and apply surprise in early-stage ideation, and how GenAI-powered visualization tools might be designed to support user-aligned, controllable, and context-aware surprise. Phase 1 consisted of semi-structured interviews with professional designers, while Phase 2 involved a participatory design workshop with graduate design students. This sequence first surfaced in-depth domain knowledge from experts and then enabled us to probe speculative system features in a more experimental, co-creative setting (see Figure 1).

3.1 Phase 1: Expert Interview

3.1.1 Participants. To better understand how surprise enhances creative design in the early stages of fashion design, and how to use GenAI to generate surprise and facilitate this process, we recruited experienced fashion designers to ensure they were familiar with and in control of the design process. This helped us avoid confounds introduced by novice workflows or inefficient practices that might distort their perceptions of GenAI. We conducted semi-structured interviews [81] with 20 professional fashion designers (Mean working experience=11.15 years, SD=6.95, 25% male, 75% female) from Asia, Europe, and the Americas, including eight workers for large employers (more than 5,000 employees)(See Table 2). This

Gender/Country		Age Group		Profession		Company Size		Years in Industry		GenAI Experience (Level)	
Male	5 (25%)	18 to 19 yrs	0 (0%)	Fashion Designer	6 (30%)	<100	5 (25%)	0-4	1 (5%)	Non-user	4 (20%)
Female	15 (75%)	20 to 29 yrs	2 (10%)	Fashion Chief Designer	1 (5%)	101-500	2 (10%)	5-9	9 (45%)	Novice	9 45%)
		30 to 39 yrs	13 (65%)	Fashion Design Manager	5 (25%)	501-1,000	5 (25%)	10-14	5 (25%)	Intermediate	6 (30%)
Asia	14 (70%)	40 to 49 yrs	4 (20%)	Fashion Design Director	1 (5%)	1,001-5,000	0 (0%)	15-19	2 (10%)	Expert	1 (5%)
Americas	2 (10%)	50 to 59 yrs	1 (5%)	Founder	1 (5%)	5,001-10,000	5 (25%)	20-24	2 (10%)		
Europe	4 (20%)			Fashion Technician	1 (5%)	>10,001	3 (15%)	25-29	0 (0%)		
		Mean	36.35	3D Artist/3D Fashion Designer	2 (10%)			30-34	1 (5%)		
		STD	6.60	3D Fashion Specialist/Expert	3 (15%)						
								Mean	11.15		
								STD	6.95		

Table 2: Demographic information of participants in the Expert Interviews

sample size aligns with saturation points in sociological research [46] and HCI standards for remote interviews [10, 74]. Participants were recruited via snowball and online methods through LinkedIn[1] and Reddit[2]. They were screened to confirm they were actively employed fashion designers.

3.1.2 Interview Protocol. Prior to interviews, participants reviewed and agreed to the university-approved consent form. Participants could preview the interview questions to assess their comfort with the study topics. Demographic and professional information was collected, including name, gender, age, location, educational background, professional experience, and prior AI exposure before the formal interview. Each interview began with a brief introduction (5 minutes), followed by a semi-structured format divided into three sections: (1) inspiration sources in professional practice, (2) conceptualization and identification of unexpected elements, and (3) AI's potential to generate surprising outcomes. Sessions lasted about one hour on average, and participants received a $25 Amazon e-gift card as compensation. The interview question lists and participants' transcription example are provided in the supplementary materials.

3.1.3 Data Collection. All interviews were conducted via Zoom[3]. With the participants' permission, we recorded each interview. We collected 20 audio recordings totaling 1,172 minutes (M=58.6 minutes) and 119,276 words of transcription (Mean=5963.8). The recordings spanned four different languages: English, Chinese, Korean, and Russian. We used OpenAI Whisper[4] to transcribe all audio into text. Subsequently, we used ChatGPT[5] to translate to English. Three bilingual researchers manually checked and corrected the translations to ensure cross-language consistency.

3.1.4 Data Analysis. The first and second authors of this paper analyzed the data using Atlas.ti[6]. They reviewed each interview together to ensure they did not overlook important details about surprise usage in fashion design and the use of GenAI. First, they discussed key elements based on the question lists: sources of inspiration, experiences with unexpected elements, and the role of GenAI in creating surprises. Then, they performed an open-ended, iterative thematic analysis [15] of the interviews, identifying significant codes from a few interviews, refining these codes as they analyzed more interviews, and later shifting to a deductive approach [134] once they developed a more structured codebook. Using this codebook, they organized the codes into relevant themes. We identified 84 codes across six themes, as shown in our codebook in the supplementary materials, which provides detailed descriptions of the codes and themes beyond those in this paper.



Figure 2: Example design task card used in the workshop.

3.2 Phase 2: Design Workshop

Based on the findings of Phase 1, we extracted the key challenges and needs of fashion designers using GenAI and compiled them into 4 design briefs, comprising 12 design tasks (see Figure 2). We conducted a design workshop to explore translating these insights into GenAI tools that help designers generate creative, unexpected images.

3.2.1 Participants. We recruited 12 graduate design students (W1-12, mean age=29.42, 25% male, 75% female). Ten had undergraduate design degrees, two had engineering degrees (See Table 3). Although they were not professional designers, all participants had taken courses that used GenAI to support or design creative tools, which positioned them well for speculative ideation about future systems. The aim of the study is to identify GenAI features that surprise fashion designers, supporting the development of fashion AI tools.

3.2.2 Workshop Procedures. This workshop was designed to create an AI tool that helps fashion designers generate creative concepts using unexpected images. Demographic and professional information was collected, including gender, age, location, educational background, and prior UI/UX experience, before the formal workshop. During the workshop, the 12 participants were first divided into four groups. Each graduate student needed to choose one design task

[1]https://www.linkedin.com/
[2]https://www.reddit.com/
[3]https://www.zoom.com/
[4]https://openai.com/index/whisper/
[5]https://chatgpt.com/
[6]https://atlasti.com/

Gender		Country		Age Group		Major		Pior Experience in UI/UX	
Male	3 (25%)	Korea	10 (83.3%)	18 to 19 years	0 (0%)	Design	4 (33.3%)	Yes	8 (66.7%)
Female	9 (75%)	Uzbekistan	1 (8.3%)	20 to 24 years	1 (8.3%)	Product/Industrial Design	3 (25%)	No	1 (8.3%)
		Mexico	1 (8.3%)	25 to 29 years	7 (58.3%)	Service design	3 (25%)	Maybe	3 (25%)
				30 to 34 years	3 (25%)	Mechanical Engineering	2 (16.7%)		
				35 to 39 years	1 (8.3%)				
				Mean	29.42				
				STD	4.37				

Table 3: Demographic information of participants in the Design Workshop

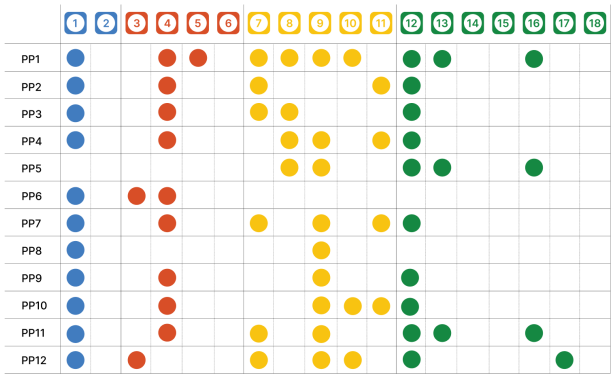
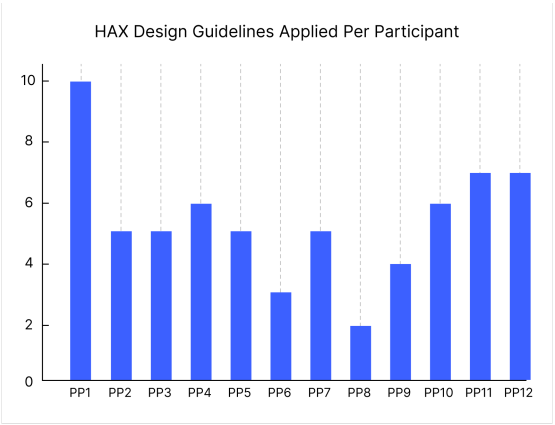


Figure 3: Consensus results from applying HAX toolkit to 12 wireframe concepts using a voting spreadsheet.



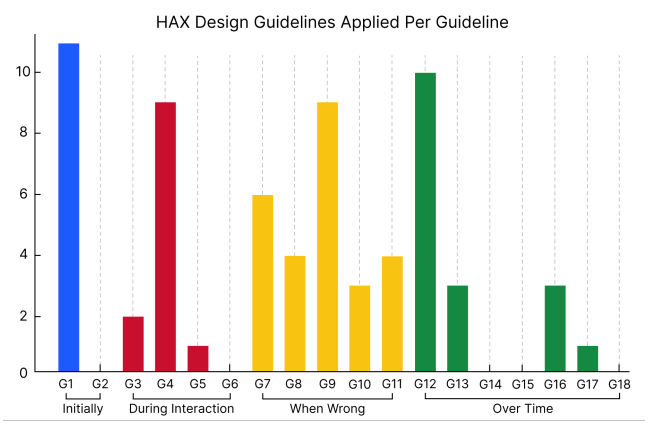


Figure 4: Distribution of HAX toolkit across participants and guideline categories in 12 wireframe concepts.

(see Figure 2). Group members could discuss within their groups to find more effective solutions. Each participant was asked to generate at least five functions or features ideas. Then, they selected the best idea within the group for each person. Afterward, participants refined the selected ideas by creating wireframes. In the end, all 12 participants presented their sketches and conducted a group-wide vote. The whole workshop lasted approximately three hours. Upon completion, each participant filled out a post-workshop survey to provide feedback and received $20 as a token of appreciation for their participation.

3.2.3 Data Collection. We used four GoPro cameras to record each team's work after they were divided into four groups, as well as the final presentation section. In total, we collected 70 rough ideas and 12 wire-frame sketch concepts from 12 participants, 301.65 minutes of video recordings, and 27.20 minutes of audio recordings for idea presentation.

3.2.4 Data Analysis. To analyze recordings and videos from the workshop, the first, second, and third authors conducted a deductive coding [31, 117] based on the HAX toolkit [141]. Before analysis, all coders participated in a calibration session to collaboratively review the HAX categories and agree on operational definitions tailored to user-facing AI products in fashion design. Afterwards, each coder independently examined the 12 wireframe sketch concepts and coded each interface element against HAX categories (see Figure 3). This involved selecting the appropriate HAX categories for each discrete interface element in the sketch. We used a shared spreadsheet to record results. After independent voting, the coders held a consensus meeting to discuss disagreements and reconcile category assignments. The reconciled votes formed the basis for the thematic grouping. The analysis resulted in four major themes: (1) Initially, (2) During interaction, (3) When wrong, and (4) Overtime (see Figure 4). After determining the theme to which each wire-frame sketch concept belonged, they further categorized the objects initiated by the instructions in each concept

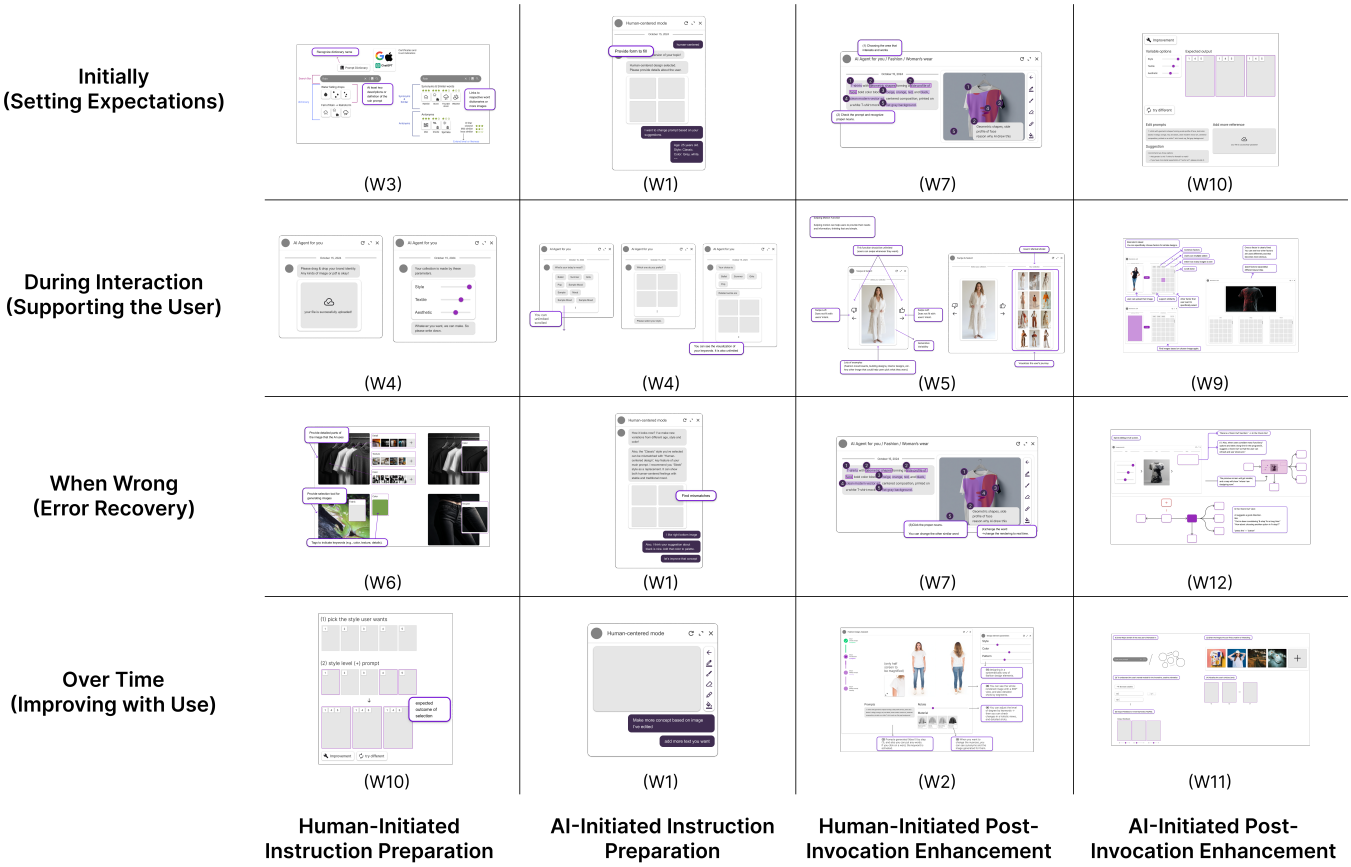


Figure 5: Overview of 12 wireframe concepts organized by: (1) setting expectations initially, (2) supporting users during interaction, (3) recovering from errors when they occur, and (4) improving with use over time (Each participant's high fidelity wireframe is provided in the supplementary materials).

according to the GenAI definition framework presented [123]. The analysis resulted in four major initiated modes: (1) Human-Initiated Instruction Preparation, (2)AI-Initiated Instruction Preparation, (3) Human-Initiated Post-Invocation Enhancement, and (4) AI-Initiated Post-Invocation Enhancement. This led to a categorization based on the interaction time and interaction object, as shown in Figure 5. It also reflected how participants' ideas mapped to themes from the expert interviews, which refined our design blueprint accordingly.

4 Results

We present results aligned with our three research questions. Section 4.1 and 4.2 (RQ1) examines how designers conceptualize and strategically utilize surprise in early fashion ideation. Section 4.3 (RQ2) analyzes how the GenAI-powered visualization tools both enable and misalign with these surprise practices. Section 4.4 (RQ3) consolidates these insights into system features that aim to support user-aligned, controllable, and context-aware surprise in GenAI-powered visualization tools. Our interview results show a design opportunity: tools could make surprise itself an adjustable parameter.

4.1 Surprise in Early-stage Fashion Ideation

We adopt the view of surprise as a discrepancy between expectation and observed reality [3]. Here, we focus on the conditions that give rise to such discrepancies in early-stage fashion ideation. Our interview data address three aspects of surprise: (1) what content and where fashion designers find surprise and (2) how fashion designers elicit surprise.

4.1.1 Surprise as an entry point: What content and where fashion designers find surprise. Prior work shows that architecture, nature, art, music, or even science fiction can be unexpected sources of inspiration [92]. Our data extends this by examining how surprise functions as an initial trigger for exploration, distinguishing between physical and digital sources of surprise, and specifically understanding where and what content designers draw inspiration for fashion design.

Physical surprise emerges from the properties of materials, construction, and embodied craft processes. For example, participants described being surprised by a low-rider car (P12), the structural complexity of sleeves (P15), traditional dressmaking techniques (P4), or the woven behavior of knit structures (P12).

> *P12: "I do like how these low riders are in stance where the car looks like it can scrape the ground at any moment [...] I like the proportions of this car, and this is another one where the construction and the materiality just multiple different levels of interpreting that inspiration."*

In contrast, digital surprises are often conceptual and perceptual, such as new color palettes, unexpected stylistic blends, or visually striking effects that catch the designer's attention and expand their creative frame. In particular, inspiration sourced through platforms for associative searches; browsing images enables divergent searches, a capability that doesn't exist when observing a single work or object.

4.1.2 Surprise as strategies: How fashion designers elicit surprise. Beyond serving as a entry point, surprise also functions as a strategy in fashion designers' early-stage workflows. For example, designers use visual-tactile incongruity and unexpected material combinations to elicit surprise [107]. Our data confirm the use of methods such as moodboards and 3D modeling, and further clarify how designers weigh surprise against feasibility.

Moodboard and 3D modeling. Participants noted that in the early stages, mood boards and concept explorations help translate unexpected inspirations into coherent visual identities. In the digital age, customers rely more on visuals since they can't physically touch products. To achieve this, our participants stated that experimentation with 3D tools or new materials often generates unexpected results.

> *P12: "There are so many surprising, unexpected instances in these design processes because I just like to play and see what kind of work it produces as I go in 3D. And that's kind of what I really enjoy about this texturing process."*

However, not all surprises are practical for production. In 3D tools, details that ignore fabric properties or pattern constraints may never be produced. During manufacturing, such surprises must be evaluated, adjusted, and made functional before they can become valuable design components.

Emotional storytelling. Research reveals that surprise, when combined with emotional storytelling, strengthens connections between products and users, spreading through social engagement and cultural discourse [22]. For example, P16 noted that, once a style is subconsciously accepted, people adopt aesthetically aligned products, which can fuel viral photos and videos on social media. For ready-to-wear, emotional resonance and narrative often matter more than overtly highlighting differences.

> *P9: "Look at luxury brands. If you take away their logo, you're looking at their artistic expression. Sometimes, when you see a style without knowing the brand, it's about the art of expression [...] they add a lot of value."*

Balance and feasibility. High-end fashion brands must balance innovation with market expectations. Successful designs balance novelty with typicality, where novelty can be considered a key feature of surprise, and typicality can be viewed as familiarity [45].

> *P2: "If you want to launch a product line [...] it needs to have a very beautiful, unexpected design point within the average level."*

Taken together, these forms of surprise have traditionally emerged through designers' interactions with materials, references, and embodied practices. However, when GenAI enters early-stage ideation, surprise is no longer only encountered through physical or experiential means. Instead, it is increasingly mediated through algorithmic recombination and data-driven inference. This shift gives rise to two distinct modes of surprise in GenAI-supported workflows, which we discuss in the next section.

4.2 Two Modes of GenAI-driven Surprise

We identify two complementary forms of surprise facilitated by GenAI: Internal Surprise (IS) and External Surprise (ES). Both forms influence how initiative is negotiated between humans and machines during the creative process. IS challenges designers' cognitive frameworks and implicit assumptions. It functions primarily at the individual level, occurring when GenAI introduces concepts beyond a designer's established territory. For example, participants in our study highlighted that AI can synthesize knowledge about hard-to-access domains, such as deep-ocean environments or dense ecosystems (P1, P15, P16), and present ideas that exceed human familiarity. Designers may temporarily hand over initiative to AI to exploit this knowledge asymmetry [124], particularly when dealing with cultural elements outside their primary framework.

> *P1: "I give ChatGPT a creative idea, like needing a lace pattern for an ocean creature series [...] ChatGPT helps me with additional knowledge and achieves a very refined effect. Though you can't look at it in detail, it provides a basic framework well."*

IS also arises when GenAI disrupts entrenched design thinking by proposing unconventional solutions to familiar problems. For example, participants believed that GenAI's works contained many designs that broke the rules and faced application. By momentarily allowing AI to lead, designers can overcome creative fixations, but they must eventually reclaim initiative to refine and contextualize these surprises.

ES emerges at the market and cultural levels, revealing opportunities that may not be immediately evident to designers. ES occurs when GenAI's data-driven analysis detects underexplored niches, such as "low-heeled dancing shoes" mentioned by P20, or "size-free underwear," mentioned by P2 and P19. These ideas began as blank spots in the domestic market, but designers recognized their potential and they later gained commercial traction.

> *P19: "About three years ago, with the sudden rise in popularity of no-size bras, it completely changed everyone's purchasing habits, even though I used to design those beautiful lace and embroidered bras with underwire."*

ES addresses collective blind spots rather than individual cognitive limits. However, because AI alone cannot guarantee functionally optimal or commercially viable designs, individual judgment becomes crucial in this context. Therefore, designers must reassert

initiative after receiving such surprises, developing or refining information to make it actionable, and transforming raw AI outputs into meaningful innovations.

Taken together, these stimuli provides designers with unexpected raw materials, but surprise alone is not enough for effective design. Designers must work with these stimuli, especially when GenAI enters the workflow, as it introduces new types of digital surprise. We next examine how fashion designers face various capabilities and challenges of GenAI in practice.

4.3 Capabilities and Limitations of Current GenAI Tools in Supporting Surprise-driven Creativity

Based on insights from 20 professional fashion designers, we examined how current GenAI visualization tools support and constrain surprise-driven creativity. Sixteen participants had used AI in their design work. Six reported company support for GenAI; three used proprietary tools, three used commercial systems such as ChatGPT and Midjourney, and one regularly relied on Stable Diffusion. These perspectives highlight how GenAI aligns with or misaligns with surprise-driven early-ideation tasks.

4.3.1 Capabilities of GenAI. **Capability 1 - Speed Visualization.** Traditionally, fashion designers rely on rough sketches, conceptual outlines, or 3D models for visualization. Tasks that once required significant 3D development time can now be streamlined with AI. Designers can quickly paint, render, and apply realistic textures to create these contextual images.

> *P12: "In terms of visualizing things, (GenAI) expedites so much that might have taken so long to do in 3D. That was my main advantage that I wanted to take in, especially because I wanted to create these contextual images."*

This capability is particularly useful in the early stages of exploration, where fast iterations matter more than perfect outputs. In this context, refer to images that capture many of the details and precise design elements that would be present in a final product. Traditional sketches are intentionally general and concise, whereas GenAI-generated images often contain richer detail and contextual cues.

Capability 2 - Design Integration. AI also excels at collecting and organizing visual and conceptual data, which is crucial when surveying a broad landscape of references. Participants noted that web-based platforms such as POP already use AI to generate extensive pattern libraries and integrated designs. While these outputs are not always production-ready, designers adapt or reinterpret them to fit production constraints.

> *P19: "Last season, I was working on integrated designs, and the advantage of integrated designs is that the structure can be highly diverse. There's now a dedicated section with runway images on POP. They have many AI-generated patterns for integrated designs, which can provide a lot of inspiration."*

GenAI has a clear advantage in gathering information and suggesting possible directions [153], while designers excel in implementation and filtering ideas for feasibility [13]. In early ideation, designers treat AI-generated surprises as raw material rather than finished products. AI provides modular inspiration, while fashion designers selectively recombine these elements into coherent concepts.

Capability 3 - Design Convey. Another strength lies in facilitating communication and emotional resonance. Even imperfect AI-generated images help teams reach a shared understanding of design intentions. They are particularly valuable for communicating design intent, surprising elements, and narrative, making it more valuable in the early design process [39, 64].

> *P10: "The commercial team doesn't have too much imagination. Designers have imagination, so from a 2D sketch, they cannot imagine what this design might look like, but the commercial team doesn't have as much imagination as designers do. So I think tools like this can really help see the product better."*

Moreover, lifelike avatars generated by AI help clients better visualize the final product, making AI valuable in both product development and marketing. This approach reduces reliance on costly physical prototypes, allowing designers to gauge reactions before production.

Capability 4 - "Wow" Creation. Designers also increasingly turn to AI to create high-impact visuals and unconventional effects, in part because its conceptual range can exceed that of a single human designer. This creative freedom, however, exists in tension with AI's limited cultural understanding. While AI can generate new combinations, it also produces outputs that lack cultural sensitivity or symbolic coherence [8, 109] and amplify cultural biases present in their training data [115]. Still, these "wow" moments can stimulate experimentation with new materials, colors, and forms. This is particularly evident when working with garment samples or prototypes, where it is difficult to deconstruct and make major changes manually.

> *P17: "If this blouse, for example, if you have it in real, you can't just take it apart very easily and make major changes and then put it back together, and then it's fine because you can't add to what isn't there [...] And that's where it changes between the physical and the digital process."*

By permitting "destructive" edits, AI supports larger, riskier design changes that may lead to innovation. For many designers, this is where AI adds the most value in practical application.

4.3.2 Challenges of GenAI. Despite these strengths, participants highlighted several challenges that limit GenAI's effectiveness in fostering surprise-driven creativity.

Challenge 1 - Lack of specificity. Participants noted that different tools have distinct image libraries, leading to variations including the designer's style, brand-specific vocabulary, and material logic. As a result, the surprise that designers intend often does not match the surprise produced by the system. In fashion design, a general concept is often needed; without targeted prompts, results

can seem like "blind box" outcomes that are unexpected and useless, requiring extra refinement. Significantly, repeated use doesn't necessarily improve system accuracy or alignment with a designer's intent.

> *P4: "I need to build a relationship with it (AI) over time for it to understand me better. But we don't have much time to nurture it due to heavy workloads. Ideally, it would analyze my language and app usage habits to adapt itself, rather than requiring my active input."*

This reflects designers' expectations that the system would adapt to their style through passive learning, rather than relying solely on active user input.

Challenge 2 - Limited understanding. AI struggles with abstract or symbolic concepts, especially when asked to express specific metaphors across different cultural contexts. For example, P2 complained that ChatGPT struggles to recognize that a banana is used as a metaphor for anxiety, because the pronunciation of "banana" in Chinese sounds similar to the word for "anxiety". Furthermore, participants criticized AI for style replication rather than a comprehensive understanding of styles and the purposeful creation of new works.

> *P11: "Really, if you give it images, you have to give it a lot of images. If you give it too few, the images it generates will be very similar to what you gave it. I want innovation, not just another version of the same design."*

Although GenAI can blend and migrate styles, the underlying logic is not understood and is patched together or referenced [144].

Challenge 3 - Limited association & expansion. Another problem is that it lacks the ability to expand on partial ideas and make associative leaps. For example, once designers explain an element, they can immediately develop it into a workable project and create prototypes. However, AI requires a clear structure and chosen elements to integrate, followed by another step to produce a deployable design.

> *P2: "Once I explain an element, a designer can immediately expand it into a workable project and arrange prototypes. AI needs a clear structure and selected elements to integrate, followed by another step to create an implementable design."*

While GenAI can extend or generate images based on existing data patterns, participants noted that the work of fashion designers involves "interactions between reality and images in terms of product realization." In short, GenAI's expansion is at the image level, while fashion designers are about integrating and validating ideas into feasible designs, from concept to reality.

Challenge 4 - Lack of iterative consistency. For many participants, GenAI functioned more like an interactive search engine than a system capable of sustained creative thinking. It struggles to generate ideas autonomously or maintain continuity across iterations. Without explicit user input, sequences lack structure, making it difficult to develop surprising ideas into workable designs.

> *P2: "I've found that in current AI systems, if I provide a clear element and ask for structural transmission, it takes longer to achieve a workable design."*

In summary, while GenAI accelerates visualization, enhances communication, and offers bold sources of inspiration, its dependence on detailed input, lack of independent reasoning, and instability limit its ability to fully support surprise-driven creativity.

4.4 System features to Enable User-aligned, Controllable, and Context-aware Surprise

We translated the challenges into the workshop and asked which system features such tools should offer to support user-aligned, controllable, and context-aware surprise during early-stage fashion ideation (RQ3). Using thematic analysis of participants' wireframes (Section 3.2.4; Figure 5), we distilled four key stages that call for specific system support: (1) setting expectations initially, (2) supporting users during interaction, (3) recovering from errors when they occur, and (4) improving with use over time.

*4.4.1 Initially Stage (Setting Expectations to support **Challenge 1 - Lack of specificity**).* Fashion designers can translate surprise concepts into tangible outputs through technical terminology and visual conventions [136], but GenAI often struggles with this fashion technical terminology, leading to misaligned surprises. Participants noted that current GenAI tools lack the specific design language of the fashion industry, which limits the system's ability to translate design intuition into artifact knowledge about garment construction, such as structure, volume, and geometry [111].

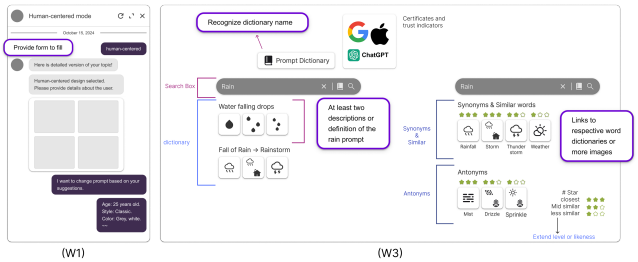


Figure 6: Wireframe examples demonstrating expectation-setting features (W1&W3).

To address this limitation, setting clear expectations in early human-AI collaboration is crucial. Participants proposed improving communication with GenAI by teaching it to understand and process design language. Research suggests that generating unexpected and valuable ideas requires the AI to first grasp what the user knows and intends to achieve [104]. Design students offered several strategies, for example, W1 proposed integrating multi-language support [94], and goal-setting features through a chatbot interface that guides users in articulating detailed prompts. W3 suggested a dictionary-like approach to clarify text inputs and explore synonyms, aligning with solutions such as "DictPrompt" [11], which combines definitions and examples to build tailored prompt templates (see Figure 6). Other students (e.g., W10, W12) recommended features that display variations, zoom-out perspectives, and directional suggestions, while W7 emphasized tutorials and contextual explanations to help users understand the impact of their inputs.

In summary, effective communication at the initial stage relies on recognizing, editing, and exploring the semantics of design language, supported by structured text and visuals. Structured prompting and definition-based clarification can be implemented via retrieval-augmented generation (RAG) [73] and contrastive decoding techniques [60, 75]. It helps models better interpret domain-specific language [142], reduces misunderstanding of designer prompts [150], thereby improving semantic precision and lowering user cognitive load [159].

*4.4.2 During Interaction (Supporting the User to support **Challenge 2 - Limited understanding**).* Parameter controls and feedback loops can help steer surprise toward a designer's intentions, rather than leaving it as random or irrelevant variation. Design language serves as an interpretation of design intuition [136], both of which are abstract. The ultimate goal is to achieve a tangible and well-constructed design outcome. When prompted with abstract terms, GenAI models tend to generate generic or poor-quality results [148]. Even with prompt engineering, visualizing abstract concepts in concrete forms remains challenging [78].



Figure 7: Wireframe example (W4) illustrating parameterized control and real-time feedback.

To address this limitation, design students proposed interface features that allow users to specify detailed parameters, such as color, shape, texture, text, or mood board (W4, W6, W8) (see Figure 7). The user's text prompt can be parsed into semantic units, which are then mapped to predefined attribute datasets. By conditioning the model with this augmented information, the system can reduce ambiguity when mapping prompts into the model's latent space [118].

Beyond input specification, real-time feedback emerged as an essential mechanism for supporting user interaction. Consistency checks and intuitive feedback loops, such as W5's swiping mechanism, where left indicates 'not fit' and right indicates 'fit', simplify the refinement process and allow designers to iteratively improve results [71]. Furthermore, Design students emphasized that explaining generated results improves understanding and empowers users to make informed edits, thereby strengthening support for micromanagement and creative control (W4, W7). Achieving this requires more interpretable AI models and clear explanations of how outputs are produced, and to trust and develop upon the system's contributions [111, 126]. However, fine-grained semantic control and real-time explainability remain technically challenging, as current mainstream T2I models lack stable internal representations for abstract concepts.

*4.4.3 When Wrong (Error Recovery to support **Challenge 3 - Limited association & expansion**).* GenAI outputs converge toward similar patterns due to shared training data sources [1, 160]. This homogenization limits diversity and stifles the novelty of surprise. To counteract this effect, systems must actively promote designer engagement, encouraging users to contribute details themselves and focus on earlier stages of the design, rather than relying solely on pixel-level inpainting [78]. Our findings suggest that meaningful and acceptable surprise ideas are more likely to emerge when designers calibrate the system before final output generation rather than only modifying results afterward.

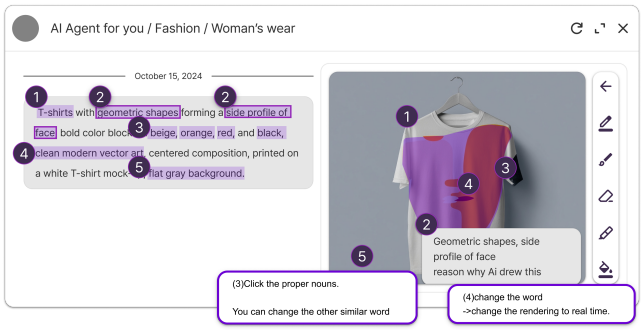


Figure 8: Wireframe example (W7) illustrating prompt precision improvements enabled by user-driven refinements.

For example, W1 introduced a chatbot interface that compares and suggests based on user prompts and generated images. Techniques such as object-level extraction can be applied using segmentation techniques such as Meta's Segment Anything [65]. Additionally, participants also suggested factor-based selection mechanisms (W4, W6, W9) to refine output relevance through contextual cues, as well as long-term precision improvements enabled by user-driven refinements (W7) [135] (see Figure 8). This can be supported through context-management frameworks that maintain user intent, preferences, and interaction history across iterative workflows [102].

Finally, navigation aids were considered valuable for error recovery. W12 proposed a zoom-out view that helps users understand their current position within the design flow, reducing confusion and clarifying process orientation. In summary, by providing adaptive feedback and navigational support early in the process rather than post hoc modification of completed outputs, GenAI can redirect surprise toward designer-meaningful directions and strengthen creative agency across the workflow.

*4.4.4 Over Time (Improving with Use to support **Challenge 4 - Lack of iterative consistency**).* Participants noted they "forget" rules over time and struggle to retain surprises they made across sessions, limiting the ability to build continuity or adapt to evolving design needs. Even though GenAI remembers certain instructions within a session, it's hard to transform these into consistent fashion designs where early development typically occurs at the collection level rather than on isolated pieces [92]. To establish the kind of rapport and emotional connection that human collaborators develop through long-term engagement, design students proposed several strategies to enhance personalization and foster sustained creative partnerships at the interface or system layer.

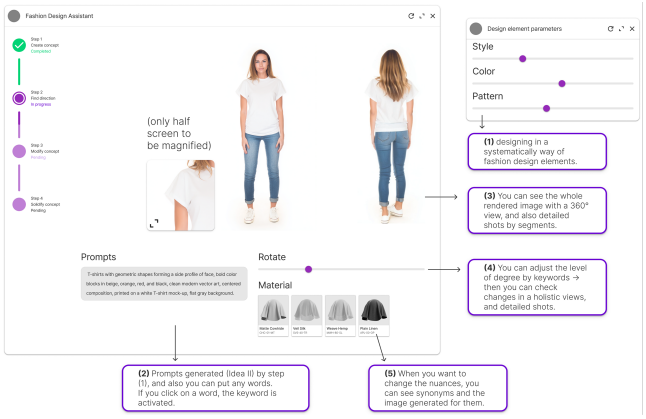


Figure 9: Wireframe example (W2) illustrating long-term workflow tracking and progressive refinement.

First, they recommended visualizing iterative workflows (W1, W11) and integrating variation mechanisms (W10) to support progressive refinement of ideas [4]. Such mechanisms allow users to explore multiple design directions and dive deeper into preferred styles. Second, design students emphasized the value of progress tracking (W2, W12) to help users understand where they are in the design process and how their ideas evolve over time (see Figure 9). For example, SketchGPT uses LangChain to construct a multi-stage and traceable design-reasoning workflow [51]. Tracking progress not only improves usability but also enhances users' awareness of their own creativity and innovation preferences [84]. Finally, incorporating continuous feedback loops was seen as essential for long-term adaptation. These loops allow systems to learn from user interactions and refine their outputs accordingly. For instance, LangGraph orchestrates multimodal agents and maintains persistent context [105] or as a workflow glue layer for continuous cross-tool editing [155, 156]. Current T2I architectures cannot adapt weights or build lasting user profiles without retraining, meaning any long-term personalization must be handled outside the model.

Improved consistency does not eliminate surprise but helps designers strategically develop surprise ideas into fashion collection. By combining iterative visualization, progress tracking, thought-pattern analysis, and feedback-driven adaptation, GenAI can move beyond "one-time" creation.

5 Discussion

Drawing on expert interviews and a design workshop, we show how "strategic surprise" in GenAI-powered visualization tools shapes designers' creative processes, decision-making dynamics, and perceptions of co-creative agency. We discuss these findings along three dimensions. Section 5.1 discusses creative initiative between humans and GenAI. Section 5.2 discusses the need for system-level coherence. Section 5.3 discusses the control in generating surprises and the ethical limits of computational creativity.

5.1 GenAI or Humans, the Initiative of Design.

Surprise arises when unexpected elements are introduced into familiar contexts, generating new aesthetic or functional possibilities [67]. While individual components may not be unique, their novel integration can produce entirely new concepts. GenAI contributes to this process by offering randomness and unpredictability that may deviate from a designer's initial vision [58], yet such deviations can open new creative avenues.

In our study, junior designers (3-5 yrs working experience) were more receptive to unexpected AI-generated ideas and tended to adopt GenAI tools more readily. Yet their limited experience, weaker design control, and underdeveloped self-directed thinking can make it difficult for them to exploit these ideas fully. Previous research has shown that experienced designers are more adept at making functional inferences from their own design sketches than novices, whose inferences are primarily perceptual [129]. A potential reason is that fashion design practice involves continuous research and absorption of environmental and experiential influences [127]. Our data also suggests that experienced designers are more cautious in their use of AI, carefully filtering generated content to avoid copyright risks and plagiarism. In summary, when GenAI produces unexpected surprises, the designer's experience ultimately shapes how these AI-generated surprises are interpreted and integrated [69].

Our participants also revealed that some traditionally emphasized skills, such as sketching, appeared less central in early design workflows when using GenAI-based visualization tools [42]. Instead, the abilities of prompt articulation, filtering AI outputs, curating composition and style, setting direction, and defining constraints will become more important. As GenAI capabilities and user interfaces improve, the entry barriers for these skills are likely to decrease. This raises the question of whether shifting from traditional, immersive making and sketching to rapid GenAI iteration will affect how fashion designers retain and comprehend design knowledge. While our data doesn't directly show this, related research suggests that rapid generation may flatten creativity [50], resulting in frustration or design fixation [56]. This is because even though inspiration from traditional sketches or moodboards is highly generalized and abstract, they possess latent possibilities that are not predetermined [138]. Heavy reliance on AI-generated suggestions could risk reducing opportunities for slow, exploratory forms of learning that traditionally support independent ideation [57–59].

As these shifts unfold, the key issue is not only what designers must learn, but also how initiative is negotiated when GenAI plays a growing role in early ideation. For example, we argue that the relationship between human initiative and computational initiative cannot be reduced to a simple dichotomy [61]. The creative process is not a linear handover of control but a fluid, dynamic exchange in which initiative alternates and overlaps [103]. While recent research proposes more nuanced scales of initiative [111, 123], we argue that tools that constrain this bidirectional flow may also constrain creativity. A model where AI merely takes over human tasks is unnatural and disrupts the creative process [78]. Instead, both internal and external surprise act as mechanisms that help designers temporarily hand over and then reclaim initiative, iterating and reinterpreting ideas not only in fashion but across the broader design field.

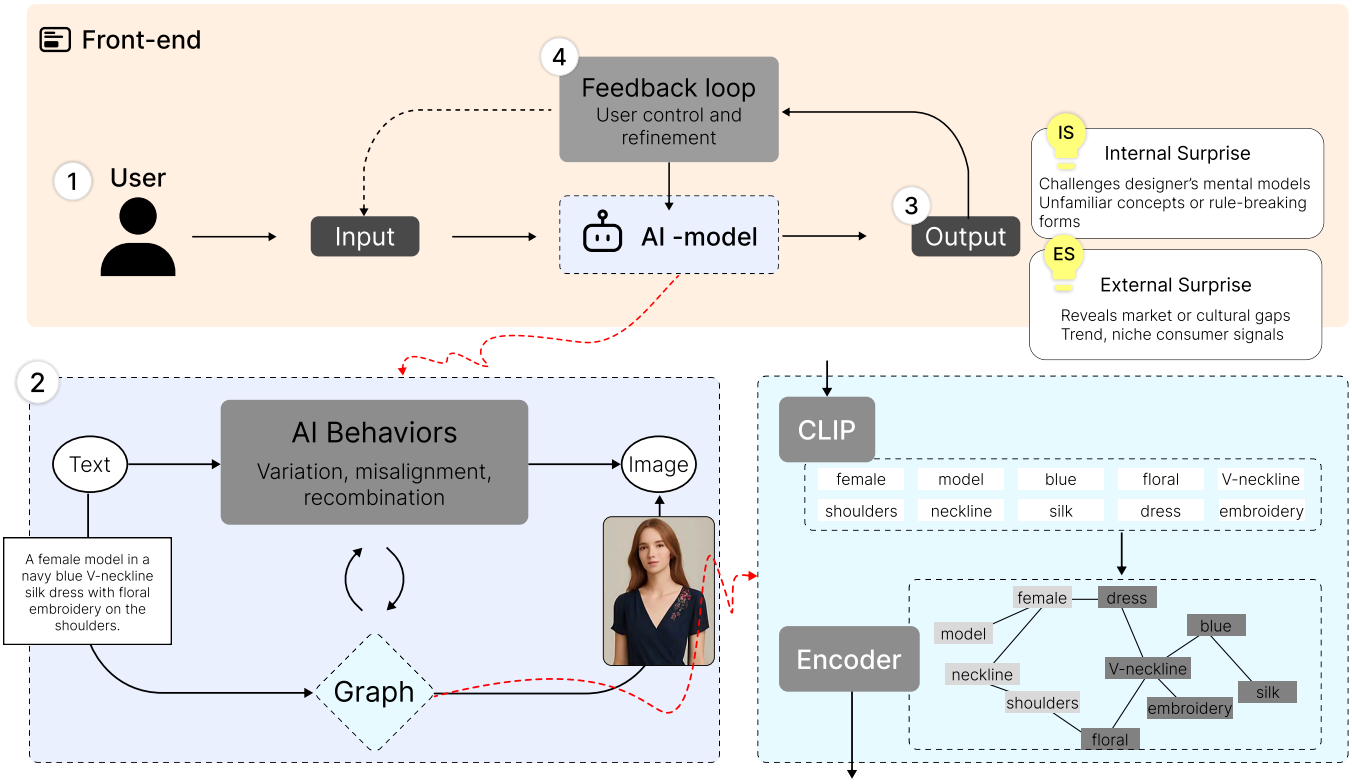


Figure 10: A methodological blueprint for strategic surprise in GenAI design, outlines four iterative stages: (1) designer inputs, (2) AI behaviors for structured surprise generation, (3) outputs categorized as Internal or External Surprise (IS/ES), and (4) feedback loops that refine both prompts and goals over time. This structure promotes a dialogic co-creative process, allowing users to explore novelty without sacrificing vision.

5.2 Enhancing Surprise Responsibly: Coherence, Control, and Context.

A key issue in using GenAI-powered visualization tools in the early stages of fashion design is significant misalignments between AI's understanding of fashion terminology and designers' own conceptual frameworks [111]. The specialized vocabulary that constitutes fashion design language varies considerably between designers, creating challenges for standardized AI interactions. The challenge becomes more acute in commercial contexts, where brands maintain distinctive design languages that constitute their identity. In response, designers expressed a clear desire for granular control during style fusion, including mechanisms that capture how they interpret professional fashion terminology and adjustable parameters for calibrating stylistic combinations [20]. Even with multilingual prompting, templates, and interface clarifications, robust handling of fashion-specific terms would require domain-adapted multimodal models.

Our findings complicate the view that control necessarily constrains creativity; in fashion design, structured selection mechanisms can enhance meaningful surprise by aligning it with collection-wide coherence [82, 127]. However, current GenAI tools fall short of supporting this process, often producing one-off novelties without regard for narrative cohesion or professional workflows. Therefore, we argue that parameter sliders, attribute tags, and swipe-based preference ranking can be built on the current embedding and similarity-ranking tools. In contrast, real-time semantic explainability remains technically challenging because diffusion models lack stable internal concept representations. Methods such as the preference-based reinforcement learning described in Section 4.3.3 can be applied to collect what ideas users favor or reject, and use these signals as rewards for fine-tuning. For example, LoRA [49] requires modest computational resources; in some cases, designers could train personalized models even on standard high-performance consumer hardware, making this approach more feasible for real-world workflows.

We also propose a methodological blueprint for strategic surprise in GenAI-powered visualization design, illustrated in Figure 10. We are not proposing a fundamentally new model, but a tool that layers interaction design and lightweight personalization on top of current architectures. It outlines four iterative stages: (1) designer inputs, (2) AI behaviors for structured surprise generation, (3) outputs categorized as Internal or External Surprise (IS/ES), and (4) feedback loops that refine both prompts and goals over time. This structure promotes a dialogic co-creative process, allowing users to explore novelty without sacrificing vision. To enable contextual

surprise, we recommend deploying fine-tuning techniques such as DreamBooth [113], Textual Inversion [33], and ControlNet [157], tools that align outputs with specific brand aesthetics, color palettes, or silhouettes. Incorporating scene-aware modules like scene graph [76, 122, 151] can further refine the semantic structure of generated images. Additionally, multilingual support through models like M2M100 [29], NLLB [16], or GPT-4 ensures accessibility and relevance for global designers. The future GenAI-powered visualization tools could extract aesthetic features (e.g., tone, fabric texture, silhouette) and manually cross-reference them with textual prompts. Designers can then use these features to generate new outputs that are unexpected yet stylistically grounded. In doing so, surprise becomes not a random outcome, but a strategic, interpretable, and user-governed affordance in fashion ideation.

5.3 AI Ethics and the Limits of Computational Creativity.

5.3.1 Limits of computational creativity. Current GenAI systems remain fundamentally limited because they replicate valued aesthetic qualities from previous artistic works [26, 43, 132, 154]. In contrast, human artists do not create solely by imitating prior examples; they draw from lived experiences such as nature, sounds, relationships, discussions, and emotions [90]. While surprise can spark creativity, it is ultimately rooted in personal struggle, self-awareness, and the pursuit of authenticity, a process shaped by emotional investment, existential risk, and cultural engagement [62, 88, 97].

In human design practice, surprise serves only as an initial entry point and cannot, by itself, lead to a complete work. Fashion designers continuously refine ideas by incorporating their own feelings and insights through iterative development. By contrast, GenAI operates within algorithmic constraints, producing outputs without lived experience or emotional depth [80, 133]. Although it can leap beyond human conventions and generate ideas that defy common sense, its results often appear "too complete" [78]. Such polished outputs are ill-suited for early creative stages, where incomplete and ambiguous ideas foster experimentation and deeper inquiry. Rather than inducing uncertainty or confusion, ambiguity in design sketches is a source of creativity, as it allows reperception and reinterpretation of figures and their groupings [35, 121, 128, 129]. When designers encounter outputs that feel final, they may experience "semantic stopsigns" [140], conceptual closures that prematurely halt exploration. While AI can mimic stylistic patterns or language, it lacks the introspection and relational grounding that characterize genuine artistic creation [21, 119]. Even advances in editing techniques, such as local inpainting [116] and instruction-based image manipulation [7], cannot fully restore the natural rhythm of human interaction with evolving ideas [78]. Beyond these technical constraints, ethical perceptions further complicate AI's role in art and design.

5.3.2 Ethical risks in GenAI-driven surprise. First, privacy and data-sharing risks arise when systems rely on users' sketches, moodboards, or proprietary design materials as part of model adaptation [109]. Participants raised concerns about privacy and data-sharing risks regarding where design materials are stored and how they may be reused. Second, the generative capacity of current models relies on large-scale training datasets that copy underlying texts, sounds, and images, which raises unavoidable questions about the copyright implications of GenAI [115]. Designers questioned whether using AI-generated outputs, trained on unconsented images scraped from online fashion lookbooks, runway photography, or cultural artifacts, would inadvertently reproduce IP violations or raise authorship ambiguity [63, 83, 109]. This tension complicates the integration of AI-generated surprise into commercial contexts, where ownership and accountability must remain clear. Addressing these privacy concerns requires robust data protection mechanisms, transparent data-handling policies, and user-friendly privacy controls [109]. For instance, research suggested deduplicating data, controlling model size, modifying training labels, and applying output filters to the reproduction of copyrighted works [109, 115].

Moreover, GenAI models inherit or amplify cultural biases embedded in their training data. As observed in Figure 10, prompts invoking "female models" often composed of lighter-skinned subjects [8]. Similar tendencies appear in color symbolism, garment silhouettes, or representations of body types, which may constrain rather than expand the range of surprising design directions. These biases are not merely technical shortcomings but reflect sociotechnical imaginaries about gender, race, and beauty, reinforcing aesthetics that can marginalize or stereotype [8, 47, 63, 109]. Such embedded norms shape the kinds of surprise that AI is capable of producing, often steering designers toward culturally dominant styles instead of enabling the diverse or counter-normative inspiration that fuels innovation.

5.3.3 Implications for responsible co-creation. Prior research shows that audiences tend to favor human-created content, especially when creative intent and personal narrative are valued [9, 66]. Participants perceive AI-generated works as disrespecting human labor, since people perceive GenAI to exert less effort than humans [85]. This objection is not only about time but about the thinking, cultural sensitivity, and emotional resonance underlying human creativity. In contexts where authorship is ambiguous, this tension becomes particularly pronounced. GenAI should not be viewed as a substitute for human creativity but as a co-creative supplement. Rather than replacing human ingenuity, AI should enhance it while safeguarding authorship, interpretability, and cultural integrity in design work [133]. It requires not only technical improvements but also ethical safeguards around data use, copyright accountability, and culturally inclusive model training. Diverse and representative training data may broaden the aesthetic scope of AI systems [80], yet this alone cannot replicate the depth of human creative experience. Our research focuses specifically on the role of GenAI in generating surprising stimuli, but questions about what is lost when convenience replaces introspection, and how to preserve creative agency in an era of algorithmic imitation, remain open for further exploration. This study aims to serve as an initial step toward these discussions.

5.4 Limitations and Future Directions

We offer an empirical account of how fashion designers conceptualize and incorporate surprise during early-stage ideation. However, our findings have not yet been examined outside the fashion domain. Our contribution is therefore not a validated tool, but an

empirically grounded articulation of a design space. By examining how fashion designers engage with and respond to surprise, we identify the critical design requirements for building GenAI-powered visualization tools. This work shifts the question from whether we should design for surprise to how we might do so to support professional creative practice.

Our predominantly Asian sample limits generalizability across diverse fashion settings. Also, interviews were conducted in participants' preferred languages (English, Chinese, Korean, Russian). While bilingual researchers verified translations, some nuances may have been lost. The workshop was conducted with design students rather than industry UI/UX or GenAI practitioners, and the scope of the stimulus materials is drawn from the fashion domain. While design students are well-suited for speculative ideation and conceptual exploration of future GenAI systems, particularly in early-stage design contexts, they do not fully represent the situated constraints, tacit knowledge, and accountability structures that professional fashion designers encounter in industry practice. As a result, the concepts generated in the workshop may emphasize exploratory creativity and interaction possibilities over feasibility, workflow integration, or organizational constraints that shape real-world adoption. The blueprint has not yet been implemented in a real GenAI prototype; thus, its user impact is hypothetical. Nevertheless, we believe the proposed design blueprint is universal and not limited to clothing design. Future research should investigate whether our blueprint for strategic surprise applies to other domains such as graphic design, architecture, game development, or advertising, which also require balancing novelty, coherence, and brand consistency.

A natural next step is to build a prototype that integrates the interaction features we identified and evaluate it with a larger cohort of designers to quantitatively assess its impact on creative diversity and ideation efficiency, to understand its sustained impact on their creative process and professional practice. This would help identify how surprise as a design affordance influences long-term creative habits and professional outcomes.

6 Conclusion

In the fashion domain, GenAI can act as a collaborator by offering strategically structured and controllable forms of surprise. Through interviews and a design workshop, we identified key designer needs around meaningful surprise, initiative negotiation, and creative enhancement. We defined the concept of surprise in fashion design and characterized its appearance in creative processes. We identify current GenAI limitations, including a lack of specificity, a limited understanding of fashion terminology, restricted association capabilities, and insufficient iterative consistency. Also, we propose a methodological blueprint outlining how GenAI-powered visualization tools could support strategic surprise through four interaction stages. While these features require validation through prototype implementation with professional designers, they represent an empirically-grounded design space for supporting controllable, context-aware surprise. Our findings highlight the transformative capacity of GenAI to stimulate unexpected visual stimuli, serving as a rich source of inspiration and creativity, encouraging it from a tool for random image generation to an inspiration-driven co-creator.

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